



Module 1 Day 1: The Cybersecurity Mindset

Class Instruction

Overview

In today's class, you will familiarize yourself with the structure of the course and learn about some of the topics that will be taught. Toward the end of class, you will learn about the CIA triad and complete a group activity.

Have fun with these activities and enjoy your first day of class!

Class Objectives

By the end of class, you will be able to:

- Explain the course structure and general direction of the program.
- Recognize the high-level security strategies and tools that will be covered in class.
- Define cybersecurity as the assessment of threats and the mitigation of risk.
- Articulate a clear definition of the CIA triad and its elements.

01. Introductions

The first part of class consists of introductions by the university, student services, and instructional staff. After these introductions, students will introduce themselves.

02. Introduce CompTIA

Your instructor will introduce the CompTIA benefits that are included with this class.

03. Emerging Cyber Threats

Now it is time to talk about why cybersecurity is such a popular skill today:

- Dependence on IT systems has grown immensely in recent years.
- There are more users online, meaning more potential targets.
- Attackers are getting increasingly sophisticated and aggressive, and launching attacks with greater audacity.
- Cybercriminals are getting more organized.
- There aren't enough skilled security professionals to fill all the available jobs.

Additionally, the emerging threat market is continuously evolving with newer threats including:

- **AI-powered attacks:** Cybercriminals use artificial intelligence (AI) to automate and enhance their attack strategies. This includes everything from sophisticated phishing campaigns that can mimic natural language to AI-driven malware that adapts to the defenses it encounters in real time.
- **Deepfake technology:** Utilizing AI-generated audio and video clips that convincingly mimic individuals, attackers orchestrate phishing schemes or social engineering attacks. These deepfakes can deceive recipients into believing they are interacting with a trusted person, leading to unauthorized disclosure of sensitive information.
- **Supply chain attacks:** Hackers infiltrate a supplier or service provider's systems to gain access to their target's network. The SolarWinds Orion breach is a notable example, where malicious code was embedded in legitimate software updates, affecting thousands of businesses and government agencies.
- **Ransomware as a Service (Raas):** This is a subscription-based model where ransomware developers sell their ransomware to other criminals who then carry out attacks. This model has significantly lowered the barrier to entry for cybercriminals looking to leverage ransomware.
- **Fileless malware:** This type of malware resides solely in memory and does not write any part of its activity to the hard disk, making it difficult to detect with traditional antivirus solutions. It typically makes use of legitimate programs to execute malicious activities.

Can you think of any new emerging threats or cyber breaches that you have heard about?

Defining Cybersecurity

What do you associate with the word "cybersecurity"?

- Cybersecurity isn't really about complicated code and evil hackers, as it is so often portrayed.
- Cybersecurity, as we will understand it in this course, is about assessing threats and mitigating risks.

You will have the opportunity to assess threats and mitigate risks in an upcoming thought exercise. First, we are going to provide an overview of the course and the tools we'll use. In this section, we will provide a high-level overview of the course, including the topics covered, technical tools used, projects, and resources provided.

04. Course Overview

Course Structure

You can think of this boot camp as six sections that each focus on an important piece of the vast cybersecurity landscape:

Unit 1: Security Fundamentals

- The first two weeks of the program are largely conceptual. This week, you will learn how to think like a security professional, explore the cybersecurity career landscape, and get an introduction to certifications in the space.
- Next week, we will focus on the emerging technology of AI and how it is used in technology and cybersecurity today.

Unit 2: System Administration

- Beginning in Module 3, you will use technical lab environments to complete activities. You will get comfortable using the command line and hone your systems administration skills in the several modules that follow.

- We will cover both Linux and Windows systems, and start programming with both Bash and PowerShell. You will configure and audit servers, and harden them from malicious attacks.

Unit 3: Networks and Network Security

- Security professionals are expected to have a strong foundation in networking. In this module, we'll cover topics such as network configuration, design, ports, protocols, and data communication.
- We'll practice analyzing data packets on the wire and investigating network security attacks and hardening, and cover a variety of topics in cryptography. This module will also examine cloud virtualization and security, and you will complete your first project.

Unit 4: Offensive Security

- With a networking foundation now established, we'll explore a variety of offensive topics in security.
- We'll start with web architecture and then explore common web vulnerabilities and the hardening techniques associated with them. We'll then cover ethical hacking and penetration testing, and use tools like Metasploit.

Unit 5: Defensive Security

- We'll now examine defensive security monitoring and spend a few weeks diving into SIEM (Security Information and Event Management) with Splunk. We will set up security monitoring and create alerts, dashboards, and custom reports.
- You will gain an understanding of the incident response framework, and learn how to respond to breaches and incidents. We'll also spend time on forensics, and use tools to recover deleted data and solve a sample forensics case.

Unit 6: Review and Final Project

- At this point, we've covered a lot! It's time for some focused certification and career prep and review. We will focus primarily on the Security+ exam but also introduce other relevant exams. In the career prep material, you will sharpen your networking skills and get practice with both behavioral and technical interviewing.

Example Assignments

Let's discuss some examples of the assignments that you will complete during this course.

- Using AI tools to generate reports, explain exploits, and analyze logs.
- Linux Scavenger Hunt: Using the skills learned in Modules 3–6, search through a Linux system to discover flags.
- Networking Capture the Flag: At the end of the Networking modules, you will work in groups to investigate data packets and find a variety of flags that tie to various networking concepts.
- Pentesting Project Week: Over three days, use the skills from the prior modules to engage in a Penetration Assessment against a Web Application, a Windows Server, and a Linux Server and generate a final report.
- Master of the SOC: At the end of the SIEM modules, you will work in groups to create a custom security operations center and use the monitoring tools you set up to analyze and protect your organization from potential attacks.

Projects

These modules and assignments will culminate in four projects:

- Project 1: Linux Hardening Project
- Project 2: Offensive Security CTF: We will compete in a three-day CTF to attack a fictional organization to determine its vulnerabilities.
- Project 3: Building a Monitoring Solution: You will design a custom monitoring solution for a fictional organization, monitoring the environment during a series of simulated attacks.
- Project 4: bootCon: In the final project, you will present independently researched projects that demonstrate one of the following:
 - Exploiting a vulnerability of an Internet of Things device.
 - Developing code or a program that can complete a cybersecurity task.
 - Demonstrating how a cybersecurity tool that was not covered in class can accomplish a specific goal.

Tools We'll Use

While the first couple of modules will primarily focus on fundamental cybersecurity concepts, we will quickly shift to applying IT, networking, and cybersecurity skills in technical environments.

Starting in Module 3, we will use virtual machines to complete many tasks associated with systems administration, networking, monitoring, programming, and more.

- You will be provided access instructions for these labs through your Canvas course.
 - There are no installations required to use this lab. All that is required is a web browser!

Student Resources

Finally, you will have access to many resources for additional learning and review.

Student Canvas: This will be your main resource for the boot camp.

This is where you will access:

- Modules:
 - Introduction: A summary of the module
 - Student Guides for the daily material
 - Activity files and lab guides for the day
 - Daily Lessons: The daily objectives for each module and links to relevant resources
 - Challenge: The weekly challenge (homework) assignment
 - Reflection: An opportunity to reflect on what you have learned and how to grow that knowledge
 - Career Connection: How to tie the information learned into your professional growth
- Labs:
 - Access to each virtual machine required for specific modules will be found right after the module in Canvas. You will start seeing labs with Module 3.
- Zoom:
 - You will be able to access the recordings for the class from here.
- Xpert Learning Assistant:
 - An AI tool designed for assisting you in your class materials
- Student Guides: The following resources can all be found on the daily lesson's page:
 - Access the slide deck for the class.
 - Access the Student Guide which is your version of a lesson plan.
 - Access the Activities files that will be done in class.

Make certain you have access to these resources. Office Hours are there to assist with getting access. These resources will be referenced throughout the course, so it is important to have access early.

05. Online Learning

 Let's discuss the online nature of this boot camp and some of the tools we'll use to facilitate and optimize learning in an online environment.

This class will be collaborative and you will spend plenty of time working together on activities. The course is designed to feel like a learning community. This first week is about making you feel comfortable with one another.

Tips

- **Get to know your classmates and instructors:** You'll get more out of the course if you feel part of a shared community. Social connection can be difficult to develop through a computer screen, but there are ways to get to know your fellow classmates. Participate in class and in your cohort's Slack space (more on Slack coming up). You don't have to get too personal, but you can share your security interests and career goals.
- **Treat class time like you would a live class:** If you were sitting in a classroom, there would not be many potential distractions around you. Likewise, you should make the physical area where you sign in to class as distraction-free as possible. If you can, go to a quiet room, silence your phone, and ask others in your home to avoid distracting you while you're online.
- **Budget time for classwork:** In an online environment, it can be harder to keep track of due dates for assignments. Deliberately scheduling time throughout the week for your coursework will help. Set aside three hours, three to four times a week, for studying and doing homework. Your work won't feel as overwhelming this way, and you won't be working on assignments at the last minute!

Zoom Online Best Practices

It's okay if you are not familiar with using Zoom in a classroom environment. We will model good Zoom behavior throughout the program.

The following link highlights Zoom features like hand raising, polling, screen sharing, and breakout rooms.

- [Zoom Features](#)

To start, here is a list of Zoom best practices:

- **Always mute:** If you are not speaking, put yourself on mute.
- **Set your first and last name:** Help everyone get to know you by including your full name as your screen name. If you are using a shared computer, check the user name when you log in to class. The TAs use Zoom to take attendance.
 - If you have a preferred name, please place it in parenthesis [Example: Robert (Booby) Smith]
- **Keep your video on:** Be present during the online class. This includes showing your face. It is understandable if something comes up and you have to go off camera for a short time, but keeping the camera on is an important part of being part of this shared community. Remember that, in your professional career, cameras are often required for remote jobs, so it is a good practice to learn now.
- **Raise your hand or use Slack for questions:** Don't interrupt a lecture. Use the hand raise feature in Zoom or ask the question via Slack.
 - A TA will answer questions after or during the lecture when appropriate.
 - A TA or Instructor will also let you know if something will be addressed in a future class or should be covered during Office Hours.
- **Use headphones with microphones:** Background noise and feedback echoes can be an issue when using your computer mic and speakers. Use headphones with a microphone to improve sound quality and clarity when speaking.

Slack

Slack is an online communication tool that is like a forum, instant messenger, and email all rolled into one. It's used by countless organizations worldwide, and you'll be using it every single day for the next six months.

- We will use Slack to send code snippets during class, share important announcements, and facilitate group exercises.
- You should have received the link to your class-specific channel during orientation. Though there is a Slack web application, for this course, you should have the program installed on your machine.
-  If you do not have access to Slack, address it during Office Hours so that you can have access to all resources.

Bringing Engagement to the Online Classroom

- The instructor will not be the only person speaking! During demos and reviews of activities, they'll pause to ask questions.
 - You should submit questions via the appropriate Slack channels. Instructional staff will make sure to cover as many of them as possible at the end of a section.
 - We will be using breakout rooms regularly throughout the course, starting with a group activity after the break.
 - We will perform frequent comprehension checks throughout class using polls. Class polls provide you with an opportunity to anonymously indicate your level of understanding of an activity.
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06. Break (0:15)

07. Introduction to First Activity and Breakout Groups

In this activity, you will start to think like a cybersecurity professional.

- Remember the main tasks of cybersecurity: assessing threats and mitigating risks.
- These two tasks often require creativity and outside-the-box thinking.

For each activity, you will be provided the instructions via an activity file that you can access on GitLab.

08. Activity: Assessing Threats

- [M1.1 Activity File: Assessing Threats](#)
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09. Instructor Review: Assessing Threats

10. Instructor Do: Defining the CIA Triad

Now, we will move on to our first cybersecurity concept: the CIA triad.

The CIA triad highlights the three cornerstones of information security: **confidentiality**, **integrity**, and **availability**.

Define the three key terms in your own words: What is confidentiality? What is integrity? What is availability?

- **Confidentiality:** The state of keeping or being kept secret or private.
 - This corner of the CIA triad is all about ensuring that sensitive information does not reach unauthorized people.
 - Some examples of confidentiality attacks include uploading private photos or communications onto a forum or exposing credit card numbers online.
 - Confidentiality comes down to the principle of "need to know": Data or information should only be made available to those who need access to it.
 - Confidentiality is supported or enforced through various other measures, like encryption, authentication, etc.
 - Example - Healthcare Data Breach: A hospital experiences a cybersecurity breach where sensitive patient records, including personal health information (PHI), are illegally accessed and leaked. This breach violates the confidentiality of the patients' data, potentially leading to identity theft and violation of privacy laws like HIPAA (Health Insurance Portability and Accountability Act).
- **Integrity:** The quality of being honest, whole, or undivided.
 - The integrity of information refers to protecting information from being modified by unauthorized people.
 - Some examples of integrity attacks include intercepting money transfers and changing the amount in seemingly insignificant ways, allowing for the excess to be sent elsewhere, or altering the grades at a university to be better or worse.
 - These kinds of attacks can be prevented by using a secure hashing algorithm and process when transferring data to ensure it isn't tampered with in transit.
 - Example: Financial Transaction Tampering: In a banking system, a hacker gains access and alters the amount of money transferred in transactions. For instance, they might increase the amount being transferred from an account, stealing money by redirecting the excess to an account they control. This action compromises the integrity of the financial data, leading

to financial loss and undermining user trust in the bank's transaction systems.

- **Availability:** The quality of being able to be used or obtained.
 - Availability concerns occur when operating systems, equipment, and data are not functioning correctly and thus are not accessible by those who need it.
 - Some examples of an availability attack include cybercriminals taking down a web-connected generator to disable a critical power supply or using a denial-of-service (DoS) attack to bring down a financial service provider's website, making it impossible for clients to make transactions.
 - Creating regular backups of data is one way to maintain availability.
 - Example: DDoS Attack on Online Retailer: An online retail website is hit by a Distributed Denial of Service (DDoS) attack during a major sale event. This attack overwhelms the site's servers with massive amounts of traffic, making the website inaccessible to legitimate users and potentially leading to significant revenue loss. The availability of the site's services is compromised, affecting customer access and satisfaction.

Cybersecurity is concerned with the confidentiality, privacy, and integrity of data and data systems.

- We will revisit these terms regularly throughout the program and explain how the various topics, skills, and practices are concerned with protecting confidentiality, privacy, and integrity.

11. Activity: CIA Triad and Security Scenarios

- In this activity, you will analyze a variety of brief security scenarios and identify which element of the CIA triad (confidentiality, availability, or integrity) each situation is concerned with.
- [M1.1 Activity File: CIA Triad](#)

12. Instructor Review: CIA Triad and Security Scenarios

13. Wrapping Up and Moving Forward

Congratulations on completing the first day of this boot camp! The pace will pick up quickly, and soon you will complete assignments and projects that showcase a completely new set of skills and comprehension.

Moving Forward

Next class, we will do a deeper exploration of assessing risk and mitigating threats by evaluating specific attacks and vulnerabilities of users, web applications, servers, and databases.

Looking further ahead, on the final day of this module, we will examine cybersecurity domains, career pathways, and certifications.